The homology of a persistence fiber without finite bars

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Abstract

For a finite connected simplicial complex, consider all simplex-wise filters whose ordinary persistent homology has one infinite degree-zero bar born at time 1 and all positive-degree bars born at time 2, with no finite bars. We prove that this fiber has the same homology over $\mathbb{F}_2$ as the original complex. The fiber has a deformation retract homeomorphic to the order complex of nonempty acyclic subcomplexes of the original complex. Comparing that poset with the face poset gives the homology isomorphism and proves the corresponding Betti-number conjecture of Leygonie and Henselman-Petrusek in the connected setting.

1 The fiber

A persistence barcode records the homology of a filtration and the maps between its stages. Different filters can have the same barcode, so the topology of a barcode fiber describes ambiguity left by this summary. Leygonie and Henselman-Petrusek [4] compute such fibers on cell complexes. Their Conjecture 1, in the preceding connected setting, predicts that a particular fiber without finite bars has the Betti numbers of the underlying simplicial complex. Polyhedral descriptions and dimension bounds for multiparameter fibers are given in [2, Theorem 5.3 and Proposition 5.9]. Here the absence of finite bars permits a direct homological comparison.

Let K be a finite connected simplicial complex. Throughout, homology has coefficients in $\mathbb{F}_2$ and is ordinary and unreduced unless a tilde is present. We identify a simplicial complex with its geometric realization when writing its homology. Simplicial and singular homology agree, including for pairs [3, Theorem 2.27 and p. 153]. Write $\beta_q(K) = \dim_{\mathbb{F}_2} H_q(K;\mathbb{F}_2)$. Let D_K contain one infinite bar born at 1 in degree zero and $\beta_q(K)$ infinite bars born at 2 in degree $q \geq 1$, with no finite bars. If all positive-degree Betti numbers vanish set $m = 1$, and otherwise set $m = 2$. Put $I = [1, m+1]$.

A filter assigns a value $f(\sigma) \in I$ to every nonempty simplex $\sigma \in K$, with $f(\tau) \leq f(\sigma)$ whenever τ is a face of σ. Its sublevel complex is

$$K_t(f) = \{\sigma \in K : f(\sigma) \leq t\}.$$

Let $\mathrm{PH}(f)$ be the graded barcode of this filtration, and give

$$F_K = \{f : \mathrm{PH}(f) = D_K\}$$

the topology inherited from $I^{|K|}$, where $|K|$ here denotes the number of nonempty simplices.

Theorem 1. *For every finite connected simplicial complex K and every integer $q \geq 0$,*

$$H_q(F_K;\mathbb{F}_2) \cong H_q(K;\mathbb{F}_2).$$

In particular, $\dim_{\mathbb{F}_2} H_q(F_K;\mathbb{F}_2) = \beta_q(K)$.

The filters in this theorem assign values to all simplices. The proof uses their whole sublevel complexes. Its homological comparison uses acyclic subcomplexes, which need not be contractible; it does not assert a homotopy equivalence between F_K and K.

Assume for now that K is nonempty. Call a subcomplex *acyclic* if it is nonempty and its reduced homology over $\mathbb{F}_2$ vanishes in every degree. The barcode D_K implies that $K_t(f)$ is acyclic for $1 \le t < 2$ and that every inclusion $K_t(f) \hookrightarrow K$ induces homology isomorphisms for $t \ge 2$. In degree zero these isomorphisms agree with augmentation: the class of every vertex maps to $1 \in \mathbb{F}_2$.

Lemma 2. *The subspace*

$$G_K = \{f \in F_K : f(\sigma) \le 2 \text{ for all } \sigma \in K\}$$

is a strong deformation retract of F_K.

Proof. For $0 \le s \le 1$ define

$$h_s(x) = \begin{cases} x, & x \le 2, \\ 2 + (1-s)(x-2), & x \ge 2, \end{cases} \qquad R_s(f) = h_s \circ f.$$

This is jointly continuous and nondecreasing in x, so $R_s(f)$ is a filter. It fixes all values at most 2. For $s < 1$, the function h_s is strictly increasing and fixes 1 and 2. It only reindexes stages after 2, whose homology maps to K are already isomorphisms; after its largest value, the sublevel is K. Thus it preserves the entire persistence module up to isomorphism. At $s = 1$, the sublevels below 2 are unchanged and the sublevel at and after 2 is K, so the same barcode is obtained. Finally R_0 is the identity, $R_1(f) = \min(f, 2)$, and every R_s fixes G_K. If $m = 1$, all values are already at most 2 and this deformation is the identity. □

A filter $g : K \to [1, 2]$ belongs to G_K precisely when

$$\min_{\sigma \in K} g(\sigma) = 1, \qquad K_t(g) \text{ is acyclic for every } 1 \le t < 2. \tag{1}$$

Indeed, necessity follows from the barcode. Conversely, the degree-zero module is constant and one-dimensional from 1 onward, since inclusions between connected nonempty complexes preserve its augmentation generator. Positive-degree homology is zero before 2 and equals that of K from 2 onward, because $K_2(g) = K$. This gives exactly D_K, also when $m = 1$.

2 Chains of acyclic subcomplexes

Let $\mathcal{A}(K)$ be the finite poset of nonempty acyclic subcomplexes of K, ordered by inclusion. For a finite poset P, write ΔP for the simplicial complex whose nonempty simplices are the nonempty chains in P. We write $|\Delta P|$ for its geometric realization.

Proposition 3. *There is a homeomorphism $|\Delta \mathcal{A}(K)| \to G_K$, affine on every simplex, given in barycentric coordinates by*

$$\Phi(\lambda)(\sigma) = 2 - \sum_{A \in \mathcal{A}(K)} \lambda_A \mathbf{1}_{\{\sigma \in A\}}. \tag{2}$$

Here $\lambda_A \ge 0$, $\sum_A \lambda_A = 1$, and the positive support is a chain of subcomplexes.

Proof. The formula is continuous and gives values in $[1, 2]$. If τ is a face of σ, every subcomplex containing σ contains τ, so the value on τ is at most the value on σ.

Write the positive support as $A_1 \subsetneq \cdots \subsetneq A_r$, with coefficients $\lambda_i > 0$. Every simplex in A_1 has value 1. A simplex whose first membership is in A_j has value $2 - \sum_{i=j}^{r} \lambda_i$, and a simplex

outside A_r has value 2. The distinct sublevels below 2 are therefore exactly $A_1, \dots, A_r$. They are acyclic, so (1) puts $\Phi(\lambda)$ in G_K.

For the inverse, list the distinct values of $g \in G_K$ less than 2 as $1 = a_1 < \cdots < a_r < 2$, and put $a_{r+1} = 2$. Define

$$A_i = K_{a_i}(g), \qquad \lambda_i = a_{i+1} - a_i \quad (1 \le i \le r). \tag{3}$$

These are nonempty acyclic subcomplexes forming a strict chain, and their positive coefficients sum to 1. If $g(\sigma) = a_j$, its memberships are exactly those with $i \ge j$, and

$$2 - \sum_{i=j}^{r} (a_{i+1} - a_i) = a_j.$$

A simplex of value 2 belongs to none of these sublevels. Hence (3) recovers g through (2), and uniquely recovers the positive support and its coefficients. In particular, zero coefficients on faces of the order complex create no ambiguity. If $\max g < 2$, then $A_r = K$, which is allowed when K is acyclic; if $\max g = 2$, its final sublevel at 2 is omitted from the list.

Thus Φ is a continuous bijection from a compact finite simplicial realization to the Hausdorff subspace G_K of Euclidean space, and is a homeomorphism [5, Theorem 26.6]. □

3 The homology comparison

The needed comparison is the acyclic case of homological poset fiber theorems; see [1, author manuscript, Section 4]. We give its finite-poset proof to make explicit why acyclicity suffices.

Lemma 4. *Let $u : S \to P$ be an order-preserving map of finite nonempty posets. If*

$$\Delta\{x \in S : u(x) \le p\}$$

is nonempty and acyclic over $\mathbb{F}_2$ for every $p \in P$, then $|\Delta u|$ induces homology isomorphisms in every degree.

Proof. Form a poset M on the disjoint union $S \sqcup P$, retaining the two internal orders and declaring $x \le p$ for $x \in S$, $p \in P$ precisely when $u(x) \le p$. There are no relations from P to S. The map $\rho : M \to P$ equal to u on S and to the identity on P is order-preserving. For the inclusion $i : P \to M$ we have $\rho i = \mathrm{id}_P$ and $\mathrm{id}_M \le i\rho$ pointwise.

Comparable order maps induce homotopic realization maps. To see this for maps $a \le b$, triangulate the prism on a chain $x_0 < \cdots < x_n$ by the simplices with bottom vertices $x_0, \dots, x_j$ and top vertices $x_j, \dots, x_n$. Their images $a(x_0), \dots, a(x_j), b(x_j), \dots, b(x_n)$ are weak chains; affine extension on these simplices gives compatible maps on all prism faces. It follows that $|\Delta M|$ is homotopy equivalent to $|\Delta P|$.

Start with ΔS and adjoin the elements of P in reverse linear extension order. When p is added, all elements of P strictly above it are present and all those strictly below it are absent. Its link is

$$L_p * \Delta(P_{>p}), \qquad L_p = \Delta\{x \in S : u(x) \le p\}. \tag{4}$$

Indeed these are exactly its lower and upper elements in the current poset, and every such lower element is below every such upper element.

The join of a nonempty acyclic finite complex L with any finite complex U is acyclic. We include the convention that the complex with only the empty simplex is the identity for join. Using augmented chains, with the empty simplex in degree -1, we have

$$\widetilde{C}_n(L * U; \mathbb{F}_2) = \bigoplus_{\ell + k = n-1} \widetilde{C}_\ell(L; \mathbb{F}_2) \otimes \widetilde{C}_k(U; \mathbb{F}_2).$$

The differential is $d_L \otimes 1 + 1 \otimes d_U$ after this degree shift. The exact finite vector-space complex $\widetilde{C}_*(L;\mathbb{F}_2)$ admits a contraction h: choose complements to its boundary subspaces and invert the differential from each complement onto the preceding boundary space. Then $d_L h + h d_L = 1$. On the join chains, $H = h \otimes 1$ satisfies

$$dH + Hd = (d_L h + h d_L) \otimes 1 + 2h \otimes d_U = 1,$$

since the coefficients have characteristic two. This proves the assertion, including when the upper poset in (4) is empty.

Adding p attaches the cone on its link. The new relative chains are the augmented chains of that link shifted up one degree: deleting p gives a face in the old complex, and the remaining boundary is the link's augmented boundary. Their homology is zero, so the long exact sequence of the pair makes each attachment a homology isomorphism [3, Theorem 2.16 and pp. 117–118, 153]. Thus $\Delta S \hookrightarrow \Delta M$ induces homology isomorphisms. Composing with ρ proves the claim for u. □

Proof of Theorem 1. For nonempty K, let $S(K)$ be its face poset of nonempty simplices, and define $u : S(K) \to \mathcal{A}(K)$ by sending σ to the subcomplex of σ and all its faces. This subcomplex is a simplex, hence acyclic, and u is order-preserving. For $A \in \mathcal{A}(K)$, the condition $u(\sigma) \subseteq A$ is equivalent to $\sigma \in A$. The lower fiber in Lemma 4 is therefore the face poset of A. Its order complex is the barycentric subdivision of A, so it is nonempty and acyclic.

Lemma 4 and the barycentric subdivision of K now give

$$H_q(|\Delta\mathcal{A}(K)|;\mathbb{F}_2) \cong H_q(K;\mathbb{F}_2) \qquad (q \geq 0).$$

Proposition 3 identifies the space on the left with G_K, and Lemma 2 replaces G_K by F_K without changing its homology. This proves the asserted isomorphism for every $q \geq 0$.

If the convention for connected complexes permits an empty complex, its unique filter has empty barcode and the prescribed fiber is empty. Ordinary homology of both empty spaces is zero in every nonnegative degree, so the same conclusion holds in that case. □

AI use. OpenAI Codex was used to develop the argument, check the proof and references, and prepare this note. Independent mathematical review is requested.

References

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